NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July – Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 2: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. The principal gain from using a heuristic function in AI search is the _____.

A. elimination of the goal test.
B. reduction of the search space.
C. generation of all successor states.
D. guarantee of finding the optimal solution.
Correct Answer: (B)

Q2. Heuristics obtained as solutions to relaxed problems are generally _____ because adding new actions should not increase the solution cost.

A. consistent
B. pessimistic
C. admissible
D. inadmissible
Correct Answer: (C)

Q3. Assertion–Reason

Assertion (A): An admissible heuristic never overestimates the actual cost of reaching the nearest goal.
Reason (R): An admissible heuristic is optimistic because its estimated cost is less than or equal to the actual cost.

Choose the correct answer:
A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true, but R is false.
D. A is false, but R is true.
Correct Answer: (A)

Q4. Which of the following statements regarding A* Search is correct?

A. A* uses only the estimated cost from the current node to the goal.
B. A* uses only the path cost from the start node to the current node.

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C. A* evaluates a node using the estimated total cost, f(n) = g(n) + h(n).

D. A* expands nodes without using an evaluation function.

Correct Answer: (C)

Q5. Assertion–Reason

Assertion (A): A* Search generates an optimal solution when h(n) is admissible for tree search and consistent for graph search.

Reason (R): An admissible heuristic never overestimates the actual cost to the goal, while consistency ensures that the estimated total path cost does not decrease along a path.

Choose the correct answer:

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true, but R is false.

D. A is false, but R is true.

Correct Answer: (A)

Q6. Assertion–Reason

Assertion (A): Simulated Annealing may accept a successor state that is worse than the current state.

Reason (R): Accepting an inferior state with a certain probability may help the search escape from a local optimum.

Choose the correct answer:

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true, but R is false.

D. A is false, but R is true.

Correct Answer: (A)

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Q7 - Q10

Consider an instance of the 8-puzzle problem. We are given the state on the left side and we want to transform it into the state on the right.

Start

1	3	4
8		5
7	2	6

Goal

1	2	3
8		4
7	6	5

- A problem with fewer restrictions on the actions is called a relaxed problem.
- The cost of an optimal solution to a relaxed problem is an admissible heuristic for the original problem.
- Two admissible heuristics are: (a) Hamming Distance h_1 and (b) Manhattan Distance h_2.

Q7. Compute the Hamming Distance of the start state for the given instance, h_1(start).

A. 5
B. 7
C. 6
D. 4

Correct Answer: (A)

Explanation: Tiles 2, 3, 4, 5, 6 are out of place. So, h_1(start) = 5.

Q8. Compute the Manhattan Distance of the start state for the given instance, h_2(start).

A. 5
B. 7
C. 6
D. 4

Correct Answer: (C)

Explanation: Contributions: 2 = 2; 3 = 1; 4 = 1; 5 = 1; 6 = 1; all others = 0. So, sum = 2 + 1 + 1 + 1 + 1 = 6.

Q9. Find the number of moves required to reach the goal from the start using the Manhattan Distance heuristic.

A. 5
B. 7
C. 6
D. 4

Correct Answer: (C)

Explanation:

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Move 0:
1 3 4
8 _ 5
7 2 6

Move 1:
1 3 4
8 2 5
7 _ 6

Move 2:
1 3 4
8 2 5
7 6 _

Move 3:
1 3 4
8 2 _
7 6 5

Move 4:
1 3 _
8 2 4
7 6 5

Move 5:
1 _ 3
8 2 4
7 6 5

Move 6:
1 2 3
8 _ 4
7 6 5
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Q10. Which of the two heuristics, Hamming Distance h_1 and Manhattan Distance h_2, is more efficient?

A. Hamming Distance h_1
B. Manhattan Distance h_2
C. Both are equally good.
D. Both are equally poor.

Correct Answer: (B)

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